ResearchX

Intelligent Research Discovery & Project Similarity Platform

1. Problem & Solution

Problem: Students and researchers often struggle to discover relevant research papers, identify projects that are too similar to an existing idea, organize paper–author relationships, and locate potential research gaps. Keyword-only search can miss papers whose meaning is related but whose wording is different.

Solution: ResearchX combines a relational database with vector similarity search. Structured information about users, projects, domains, papers and authors is stored in MySQL, while semantic embeddings of project descriptions and research text are referenced in a vector database such as Qdrant. Users can search papers, analyze project similarity, receive related research results and explore structured research relationships.

2. System Workflow

1	Input	Student enters a project idea, problem statement or research topic.
2	Process	The application prepares the text and generates an embedding for semantic comparison.
3	Store	Structured records go to MySQL; vector references and embeddings go to Qdrant.
4	Search	Researchers use full-text search or semantic similarity to retrieve relevant papers and similar projects.
5	Explore	Next.js displays matches, similarity scores, research relationships and analytics.

3. Technology Stack & Architecture

Layer	Technology	Role in ResearchX
Frontend	Next.js	Research dashboard, project analyzer, semantic search and role-based interface.
SQL	MySQL	Users, projects, domains, papers, authors, reviews, audit logs and full-text search.
Vector DB	Qdrant	Project and research-paper embeddings for semantic similarity search.
AI/NLP	Embedding Model	Converts project descriptions, abstracts and research text into vectors.

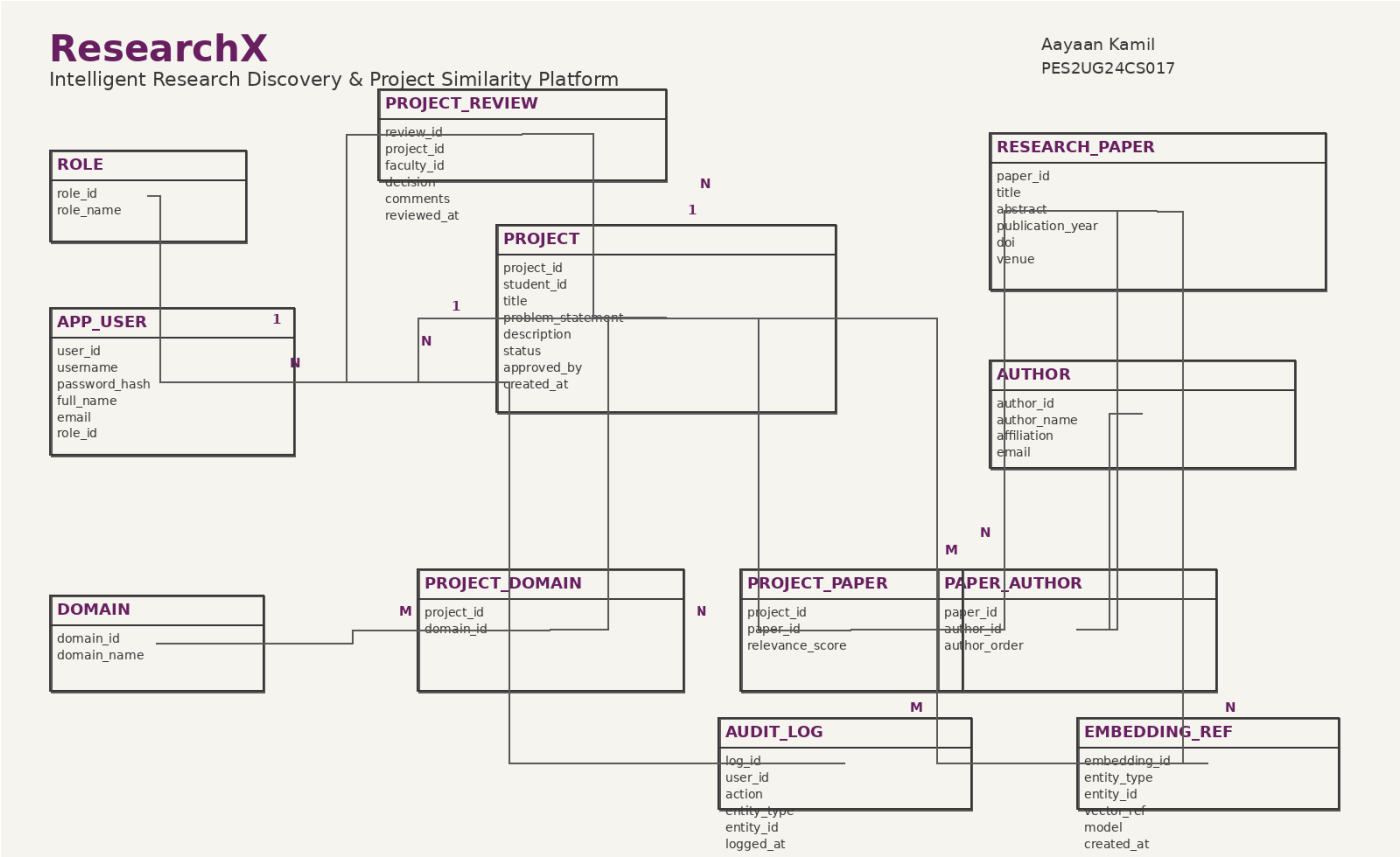
4. Key DBMS & Intelligent Features

Vector Similarity: semantic project/paper matching using embeddings. **Full-Text Search:** MySQL indexed search over research text. **Triggers:** automatically record project status changes. **Stored Procedures:** controlled project approval operations. **Transactions:** keep approval and review updates atomic. **Indexes:** speed up project and paper lookup.

5. Roles & Interface

Admin — manage users, domains and system records. **Student** — search papers, create projects and analyze similarity. **Faculty** — review projects and view research analytics.

ER Diagram



Relational schema

Relational schema

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